SATHISHKUMAR R

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Education

Vellore Institute of Technology, Chennai — **July 2025 – Expected May 2027**
M.Tech in Computer Science and Engineering — *CGPA: 9.64/10.0*

Shiv Nadar University, Chennai — **September 2021 – June 2025**
B.Tech in Computer Science and Engineering (IoT) — *CGPA: 8.63/10.0*

Experience

Intel — **July 2026 – Present**
AI Software Engineer Intern — *India*

Tata Consultancy Services (TCS) — **May 2024 – July 2024**
IoT Engineer Intern — *Chennai, India*

- Built IoT integration models for manufacturing, improving data collection efficiency by 30% via sensor deployment and edge computing; analysed pharma and automobile workflows, finding 5+ optimisations (QC accuracy +25%).

AppLever Tech — **Dec 2023 – Jan 2024**
Software Developer Intern — *Hybrid, Bangalore, India*

- Engineered a Flutter event app with geofencing and the Google Maps API, cutting registration time by 40% and improving check-in accuracy by 50%; Firebase backend supported 1,000+ concurrent users with 100% sync reliability.

ISRO Propulsion Research Complex (IPRC) — **June 2023 – July 2023**
In-Plant Trainee — *Mahendragiri, India*

- Hands-on with VLAN segmentation, firewall configuration and secure mail servers in a high-security facility.

Projects

Federated Learning Framework for NIDS | *Python, TensorFlow, Differential Privacy, Byzantine Aggregation* — **Sep 2025 – Present**

- Developing a privacy-preserving collaborative NIDS without sharing network traffic; targeting 95% accuracy and 20% fewer false positives via Byzantine-resilient aggregation and adaptive differential privacy.

Ouroboros – LLM-Enhanced SSH Honeypot | *Python, Paramiko, Flask, React, D3.js, Gemini* — **2025 – Present**

- Built an SSH honeypot in Python and Paramiko that uses Gemini 2.5 Flash to generate realistic terminal responses for unknown attacker commands, making it harder to detect via traditional fingerprinting.

FoodIndex – Surplus Food Marketplace | *MERN Stack, MongoDB Geospatial, Node-Cron, React Context* — **Jan 2026 – Feb 2026**

- Architected a hyper-local food rescue platform with reverse Dutch auction pricing: MongoDB $geoNear discovery and cron-driven repricing via a 4-factor algorithm (40% time decay, 30% stock, 20% demand, 10% urgency).

Sentinels Network Monitor | *Python, Flask, Scapy, SQLite, WebSockets* — **Dec 2025**

- Built a multi-threaded Scapy packet sniffer detecting SYN floods, UDP floods and port scans with 92% accuracy.

GHA-BFP+ – Build Failure Prediction Bot | *Python, Machine Learning, GitHub Actions, Flask* — **2025**

- Built a GitHub Actions bot using a trained ML model to flag CI/CD runs likely to fail, cutting developer wait time.

Discord Bot Ecosystem | *Python, Discord.py, MongoDB, Flask* — **June 2022 – Aug 2023**

- Created a multi-functional bot serving 6,000+ users across 20+ servers with a virtual economy and moderation tools, handling 50,000+ daily commands with 99.8% uptime on Heroku.

Technical Skills

Programming Languages: Python, JavaScript, Java, C, Dart, SQL, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Flutter
Cloud & Databases: Firebase, MongoDB, MySQL, AWS
Developer Tools: Git, GitHub, Docker, VS Code, Postman, Android Studio, Figma